Parallel Processing Concepts

Q. What is Parallel Processing? Discuss the need for parallel processing, its types, and its primary advantages in computer architecture.

> **Definition to Remember**

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1. Need for Parallel Processing

Historically, processors became faster by increasing the clock frequency. However, this approach hit physical limits (the **power wall**), causing extreme heat and power consumption.

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2. Types of Parallelism

1. **Instruction-Level Parallelism (ILP):**

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3. Advantages of Parallel Processing

- **Increased Speed / Throughput:** Executing tasks simultaneously drastically reduces total execution time.

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> **Must-Write Points (for 10 marks)**

> 1. **

> **Quick Recall**

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